Khairul Ammar Hakimi

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Education

B.A. International Business & B.S. Chemical and Biomolecular Engineering (Double Major)

Feb 2023 – Dec 2026 (expected)

Sogang University / Sogang Business School & College of Engineering · Seoul, South Korea

Converted score 99 / 100

Research Experience

Research Assistant — Clean Energy Process Laboratory (CEPL)

Sep 2026 – Present

Prof. Ung Lee · Dept. of Chemical & Biomolecular Engineering, Sogang University

- Designed an 11-sheet P&ID (Rev C) for a lab-scale direct-air-capture → reactive-capture CO_2 electrolysis → methanol-syngas loop: 7 modular skids, 20 metered interfaces, 8 control loops, 6 safety interlocks; 22 open assumptions tracked in a HOLD register, each tied to the measurement that closes it
- Built the control strategy around six root causes of absorbent pile-up: K^+ bleed ratioed to cell current, loading control from a pH/conductivity soft sensor, level-controlled water makeup
- Rebuilt the lab's ACS Catalysis (2025) interpretable-ML paper on CO_2-reduction catalyst degradation as an 18-scene interactive walkthrough, with every curve regenerated from a Butler–Volmer + limiting-current model (catalyst-xai-deck.vercel.app)

P&ID · Process control · CO_2 electrolysis · Electrochemical modelling · three.js

Undergraduate Researcher — Membrane Research Group

Mar 2024 – Dec 2025

Prof. Jong Suk Lee · Dept. of Chemical & Biomolecular Engineering, Sogang University

- Spun hollow-fibre membranes for petroleum-refining separations, then characterised them and ran permeation tests
- Reviewed literature to support the group's membrane projects

Hollow-fibre spinning · Membrane characterisation · Permeation testing

Chemical Engineering Projects

QM Property Predictor

2026

huggingface.co/spaces/taeyangkimi/solubility-predictor

- Tested whether B3LYP/6-31G* quantum features (HOMO/LUMO, dipole, Mulliken charges) improve aqueous-solubility prediction over cheap descriptors; PySCF over 200 molecules
- Measured a null result rigorously: scaffold k-fold, paired bootstrap, and noise controls that showed the one apparent gain was reproduced by Gaussian noise
- Quantified the experimental noise floor at 0.34 log units by cross-checking two independent curations, establishing what any model could achieve
- Deployed predictor returns a TreeSHAP attribution and applicability-domain warning with every prediction

Python · PySCF · RDKit · XGBoost · PyTorch

Reaction Simulator

2024 – Present

reactionsimulator.vercel.app

- Real-time concentration and conversion simulator for CSTR, PFR, and reactors in series
- Arrhenius-based temperature-dependent rate modelling and visualisation
- Turns reaction engineering and thermodynamics theory into an interactive learning tool

JavaScript · Python · Flask · Vercel

Fugacity Simulator

2024

fugacity-simulator.vercel.app

- Fugacity calculations from Peng-Robinson and van der Waals equations of state
- Automatic generation and plotting of vapour-liquid equilibrium phase diagrams

Python · NumPy · Matplotlib · Flask

Process Design Game

2024

process-design.vercel.app

- Interactive simulator for optimising distillation columns, heat exchangers, and reactors
- Scoring system giving real-time feedback on energy efficiency and economic metrics

JavaScript · Canvas API · Python

Linear Programming Diet Optimisation

2024

Academic project

- LP model producing a minimum-cost diet under nutritional constraints
- Applied engineering optimisation methods to everyday life systems

Python · SciPy · PuLP

Apple Supply Chain Analysis

2024

Academic project

- Risk analysis of semiconductor and materials supply chains with numerical bullwhip-effect simulation
- Derived data-driven SCM optimisation strategies and produced a visual report

Python · Pandas · Excel

Software Projects

LifeOS — Personal Productivity OS
2024 – Present

lifeos-wine.vercel.app

- Full-stack personal operating system unifying habits, goals, finances, and study
- Built AI-connected automation; in active use by real users

Next.js · TypeScript · PostgreSQL · Vercel

GLEAM — Student Council Governance Platform
2026 – Present

samsunggleam.vercel.app · Global Hope Scholars Student Council

- Bilingual (EN/KO) operating platform: member portal with a club → period → event hierarchy, admin governance module, club-report export renderer
- Built so the next cohort inherits a working system, not institutional memory — after PALS's constitution and handoff protocol

Next.js App Router · TypeScript · Supabase · i18n

CNN Emotion Classifier
2024

Personal project

- Trained a CNN for real-time facial expression recognition
- Built an emotion-driven UI colour adaptation system

Python · TensorFlow · OpenCV

Yummz — Student Community Platform
2026 – Present

yummzsg.vercel.app · International Student Council

- Event and community platform: RSVP flow, organiser portal, opt-in event chat with moderation and scheduled retention jobs
- Delivered to a fixed deadline ahead of the autumn planning cycle

Next.js · TypeScript · Drizzle ORM · PostgreSQL

Interpreting & Translation

Korean–Malay Investigative Interpreter (Suspect Interrogation)
Jun 2026

Incheon Airport Customs · Incheon International Airport Terminal 2

- Sole Korean–Malay consecutive interpreter for a narcotics-control law violation case
- Served under a signed interpreter's affidavit in a setting demanding high accuracy and confidentiality

Korean–English–Malay Official Interpreter
May 2026

Gyeongsangbuk-do Provincial Government / Office of Education · Gyeongsangbuk-do

- Two-way interpreting between the Vice Governor for Economic Affairs and Malaysia's former Minister of Federal Territories
- Interpreted preliminary talks for an education exchange MOU between Gyeongbuk and Selangor

Korean–English–Malay Official Interpreter
First half of 2025

Gyeongsangbuk-do Provincial Government · Gyeongsangbuk-do

- Korean–Malay simultaneous interpreting for official talks between the Governor and the Malaysian Ambassador to Korea
- Handled advanced diplomatic protocol and regional industry cooperation terminology accurately

Official & Volunteer Campus Interpreter
Feb 2023 – Present

Sogang University Office of International Affairs · Seoul

- Ongoing Korean–English and Korean–Chinese consecutive interpreting and document translation for international students

Korean–English–Malay Freelance Interpreter
Nov – Dec 2023

Supreme Prosecutors' Office of the Republic of Korea · Seoul

- Cut meeting time with overseas clients by 10–15 minutes through trilingual legal terminology interpreting
- Ran cultural briefing sessions for a visiting senior Malaysian delegation

Korean–English–Malay Freelance Interpreter
Nov – Dec 2023

Anti-Corruption & Civil Rights Commission · Daejeon

- Trilingual two-way simultaneous interpreting on finance and law for international trainees' civic education
- Translated training materials, raising trainee competency assessment scores by 12%

Korean–English–Malay Freelance Interpreter
Feb – Aug 2022

Embassy of Malaysia · Seoul

- Medical interpreting for Malaysian students at hospital visits, ensuring accurate patient–doctor communication

Korean–English–Malay Freelance Interpreter
Oct 2023 – Present

ASEZ (Save the Earth from A to Z) · Seoul

- Simultaneous interpreting across Korean, Malay, English, Chinese, and Japanese; contributed to reaching 300,000 cumulative signatures

Leadership · Scholarships · Awards

PALS Co-President
Mar 2025 – Present

Sogang University international student club PALS (Peer Assistance & Learning Support)

- Lead mentorship and exchange programmes for international students from 15+ countries
- 2024-1 Outstanding Mentor Activity Award · 2024-2 Mentor of the Semester

Samsung Global Dream Scholar
Mar 2024 – Present

Samsung Dream Scholarship Foundation — Global Hope Scholarship Award

- Selected by the foundation as a prospective global leader; full tuition plus living stipend

Sogang Alumni Excellence Scholar
Mar 2025 – Present

Sogang Alumni Scholarship Foundation (Sogang University)

- Selected as an excellence scholar in Chemical and Biomolecular Engineering (No. 25-STU-083)

Lee&Won Asian Fellowship · Leadership Excellence Award

Jan – Dec 2023

Lee&Won Foundation — Asian Fellowship Programme

- One of 18 fellows chosen from 500+ applicants; led a 17-member team spanning 15 countries
- Received the Exemplary Leadership Excellence Award (Dec 2023)

Korean Language Support — Refugee Mentoring

One-to-one voluntary support, connected through Sogang University

- Provided sustained Korean-language learning support to a Somalian refugee, translating Korean academic and social systems for an adult entering them from outside

Skills

Languages	Korean TOPIK Level 6 (265/300) · English native · Malay native · Chinese business · Japanese conversational
Interpreting	Simultaneous and consecutive — diplomatic, legal, investigative, medical, and corporate domains
Chemical Eng.	Reaction engineering, thermodynamics, process design, phase equilibria, linear programming, supply chain analysis
Data & SW	Python (NumPy · SciPy · Pandas · TensorFlow · Flask) · Next.js · TypeScript · PostgreSQL · Excel
Other	Multinational team leadership, educational content design, project management, applied AI tooling